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Palaeomagnetism of the Neoarchaean Pongola and Ventersdorp Supergroups and an appraisal of the 3.0–1.9 Ga apparent polar wander path of the Kaapvaal Craton, Southern Africa

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Abstract

The style of tectonic processes on the young Earth is a topic of intense debate. Palaeomagnetism provides the only robust tests with which to settle theoretical disputes based on the rock record that infer large scale horizontal motions between palaeocontinents. A possible apparent polar wander path (APWP) of the Kaapvaal Craton shows significant distances between adjacent palaeomagnetic poles that can be interpreted as evidence for such horizontal displacements during the Archaean-Palaeoproterozoic. There are, however, substantial uncertainties build into this APWP. Currently, the published APWP for the Kaapvaal Craton comprises eight palaeopoles between 3.0 and 1.9 Ga, including several between 2.8 and 2.7 Ma that indicate possible horizontal motions exceeding 6500 km. The aim of this contribution is to increase the pole density within this interval and in doing so to test the robustness of the Kaapvaal APWP within this time frame. More than 250 samples of basalt, diamictite and dolerite of the Neoarchaean Pongola (ca. 2.95–2.85 Ga) and Ventersdorp (ca. 2.71–2.70 Ga) Supergroups of the Kaapvaal Craton were sampled both at the surface and in an underground mine. Thermal demagnetisation revealed two magnetic components. A high temperature component is present in three Pongola formations and in two Ventersdorp formations; these may be recorders of the primary natural remanent magnetisation. Because of lack of precise geochronology and positive field tests, this magnetisation can only be constrained to be older than ca. 2.05 Ga. A medium temperature overprint in many samples probably records remagnetisation at ca. 180 Ma, during the emplacement of basalts and dolerites of the Mesozoic Karoo Large Igneous Province. This remagnetisation appears to have affected a vast area of the Kaapvaal Craton. Our new palaeopoles, together with those published previously, demonstrate that the Neoarchaean APWP of the Kaapvaal Craton remains poorly defined. Direct comparisons between the APWP of the Kaapvaal Craton with APWPs of other Archaean cratons to determine relative motions between Archaean continents, therefore, need more reliable data.
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1. Introduction

The Kaapvaal Craton of Southern Africa encompasses some of the best-preserved parts of stable continental crust of Neoarchaean age (3.0–2.5 Ga). The dominant mode of tectonism during within this period is still a topic of intense debate (e.g. Bédard et al., 2003; de Wit, 1998; Hamilton, 1998; McCall, 2003; Cawood et al., 2006). Here, we contribute to this debate using palaeomagnetism. We studied the Neoarchaean Pongola and Ventersdorp Supergroups because these successions contain abundant basalts, rocks that are

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usually good recorders of the fossil geomagnetic field at the time of their origin. The degree of metamorphism of these rocks is relatively low, generally at lower greenschist facies, which provides a reasonable chance that their original natural remanent magnetisation (NRM) is preserved. However, parts of the Kaapvaal Craton have been subjected subsequently to a number of regional Proterozoic and Phanerozoic tectonothermal events (e.g. Eglington and Armstrong, 2004; Poujol et al., 2003), including: the intrusion of the vast Bushveld Complex at ~2.057 Ga; the Vredefort Meteorite Impact at ~2.017 Ga; the Waterberg Large Igneous Province (~1.87–1.98 Ga); the Natal-Namaqua orogenic event (~1.040 Ga); and the emplacement of the Karoo Large Igneous Basalt Province (~0.182 Ga). It is uncertain to what degrees these events may have affected the original NRM, but widespread remagnetisation has been recognised across parts of the Kaapvaal Craton in previous palaeomagnetic studies (e.g. Layer et al., 1988b).

Obtaining a well-dated, reliable palaeopole requires, in part, demonstration of the primary nature of the magnetisation in a dated rock unit. Several other criteria for reliability, including knowledge of the palaeohorizontal plane at the time of magnetisation, are discussed by Van der Voo (1990). Secondary overprints also yield useful information, because secondary poles lie on, and help to define, younger parts of the APWP for the craton, and may give insight into its subsequent thermal history. Finally, there have been suggestions, based on lithostratigraphical, geochronological, and palaeomagnetic data, that the Kaapvaal and Pilbara Cratons may once have been part of the same continent called Vaalbara (e.g. Cheney, 1996; Strik et al., 2003; Wingate, 1998; Zegers et al., 1998). Here, we examine the validity of this hypothesis further.

Palaeomagnetic studies of the Kaapvaal Craton are limited and of variable quality. This mostly reflects the fact that previous studies were completed before progressive demagnetisation became a routine procedure. The Global Palaeomagnetic Database (GPMDB, see URL in references) lists just three Archaean palaeopoles for the Kaapvaal Craton between 3.2 and 2.5 Ga (Table 1). A further six (presumed) primary palaeopoles are published for the period between ca. 3.0 and 1.9 Ga (Table 1). Reconstructions of an APWP are further limited by the small number of radiometric dates that are available to accurately date the paleopoles (Table 1). Thus, a reliable APWP for this time period does not exist. Yet, these palaeomagnetic data have been used to support large degrees of rifting and high rates of drifting in the Archaean for the Kaapvaal Craton (e.g. Kröner and Layer, 1992; Layer et al., 1989). Therefore, we will here also review the quality of the previous data and compare these with our new data.

2. Geological setting

The proto-Kaapvaal Craton formed over a period of approximately 500 million years, between 3.6 and 3.1 Ga, through amalgamation of a number of tectonic blocks (de Wit et al., 1992; Eglington and Armstrong, 2004; Poujol et al., 2003; Schmitz et al., 2004). By 3.0–2.9 Ga, the craton was part of a relatively stable continent with a well formed 30–50 km thick continental crust underlain by a ca. 250–300 km thick lithospheric mantle keel (Moser et al., 2001; Stankiewicz et al., 2002; Fouch et al., 2004; Schmitz and Bowring, 2004; Schoene and Bowring, 2007). Remnants of the first major sedimentary basins that were deposited across peneplained granite-greenstone basement of this continent are represented by the siliciclastics and volcanics of the Witwatersrand-Dominion and Pongola Basins. The Pongola and Witwatersrand are contemporary deposits (Table 1), possible of the same regionally extensive basin (Fig. 1). Below we summarise their salient geological features that are also summarised in Table 1.

2.1. Pongola Supergroup (ca. 2.95–2.85 Ga)

The Neoarchaean Pongola Supergroup is exposed in an area of ca. 120 km by 275 km in the southeast of the Kaapvaal Craton, partly in Swaziland and partly in South Africa (Fig. 2). The Pongola Supergroup unconformably overlies the granite-greenstone basement of the Kaapvaal Craton (e.g. Button, 1981). The sequence comprises a maximum stratigraphic thickness of ca. 12 km of volcanics and clastic sediments, but with significant lateral variations in thickness (Burke et al., 1985). The Pongola Supergroup has been subdivided into a predominantly volcanic Nsuze Group (e.g. Armstrong et al., 1982) and a sedimentary Mozaan Group (e.g. Beukes and Cairncross, 1991, Table 1). The Nsuze Group has been interpreted to represent a rapid phase of rifting followed by deposition of the Mozaan Group as shallow marine shelf deposits during thermal subsidence of the Pongola Basin (Bickle and Eriksson, 1982; Burke et al., 1985; Beukes and Cairncross, 1991).

The entire Pongola Supergroup has been affected by greenschist-facies metamorphism (Burke et al., 1985). The Pongola Basin has experienced moderate Archean deformation. In the extreme south, the deformation is more intense because of overprinting by the 1.0–1.04 Ga Natal-Namaqua orogeny (Burke et al., 1985; Matthews, 1990).

Table 1
Summary lithostratigraphy and geochronology of the Pongola and Ventersdorp Supergroups, and other rock units relevant to this study

Supergroup	Group/ *Intrusive*	Subgroup	Formation	Dominant lithology	Pmag ref.	Age, method (reference)
Younger	*Post-Waterberg*			dolerite	a	1879-1872 Ma, U-Pb baddeleyite [1]
	Bushveld Complex			ultramafic	b*	2059 ± 1 Ma, U-Pb titanite [2]
	Postmasburg		Ongeluk	lava	c	2222 ± 13 Ma, Pb-Pb [3]
	Younger rocks			various		2642 ± 3 Ma, Pb evap.[4]
	Mbabane Pluton			granitoid	d*	2687 ± 6 Ma, $^{40}Ar/^{39}Ar$, hornblende [5]
Ventersdorp	(Pniel)		Allanridge	flood basalt	this study	(older than 2642 ± 3 Ma)
			Bothaville	sediments		
	Platberg		Rietgat	mafic volcanics		
			Makwassie	porphyry		2709 ± 4 Ma, U-Pb zircon [6]
			Goedgenoeg			
			Kameeldoorns	sediments		
	Klipriviersberg		Edenville	subaerial flood basalt		
			Loraine			
			Jeanette			
			Orkney			
			Alberton			2714 ± 8 Ma, U-Pb zircon [6]
		East Driefontein	Westonaria		this study	
			Venterspost			
	correlation uncertain		Derdepoort	flood basalt	e*	2782 ± 5 Ma, U-Pb, zircon [7]
Unconformity/ various intrusions	*Gabarone Complex*			granite	f	2783 ± 2 Ma, U-Pb zircon evap.[8]
	Modipe Gabbro			gabbro	g	> 2783 Ma
	Usushwana Complex			gabbro, pyroxenite		2875 ± 40 Ma, $^{40}Ar/^{39}Ar$, hornblende[9]/ 2871 ± 30, Sm-Nd[10]
	Post-Pongola			dolerite	this study	Age unknown
Pongola	Mozaan	Nkoneni	Ntanyana	sediments, BIF, minor volcanics		
			Gabela			
			Bongaspoort			
		Odwaleni	Kiphunyawa		this study	
			Tobolsk		this study	
			Delfkom			
		Dwaalhoek	Hlashana			
			Thalu			
			Ntombe			
			Sinqeni			< 2837 ± 5 Ma, U-Pb zircon [11]
	Nsuze	Ozwana	Mkuzane	mafic volcanics		
			Langfontein			
		Bivane	Agatha		this study	2984 ± 2.6 Ma[12], 2940 ± 22 Ma, U-Pb Zircon[10]
			White Mfolozi			
			Nhlebela		this study	
			Mantonga			(younger than 3028 ± 14 Ma)
	Older basement			granite-greenst.		3028 ± 14 Ma, Rb-Sr[13]
	Nelshoogte Pluton			granitoid	h*	3179 ± 18 Ma, $^{40}Ar/^{39}Ar$, hornblende[14]

	Felsic volcanics
	Mafic extrusive
	Mafic Intrusive
	Sediment
	Mixed lithology

Lithology (Gold and von Veh, 1995; Kent, 1980; van der Westhuizen et al., 1991; Winter, 1976). References to published palaeomagnetic data (Pmag Ref.), poles with an asterisk (*) are listed in the GPMDB: (a) Hanson et al. (2004); (b*) Hattingh (1986); (c) Evans et al. (1997); (d*) Layer et al. (1989); (e*) Wingate (1998); (f) Evans (1966); (g) Evans and McElhinny (1966); (h*) Layer et al. (1998). Geochronology: (1) Hanson et al. (2004); (2) Buick et al. (2001); (3) Cornell et al. (1996); (4) Minimum age for the younger successions (Beukes, unpubl., reference in Tinker et al., 2002); (5) Layer et al. (1989); (6) Armstrong et al. (1991); (7) Wingate (1998); (8) Grobler and Walraven (1993); (9) Layer et al. (1988a); (10) Hegner et al. (1984); (11) Minimum age from felsic sill (Gutzmer et al., 1999); (12) Hegner et al. (1993); (13) Barton et al. (1983); (14) Layer (1986) and Layer et al. (1998).

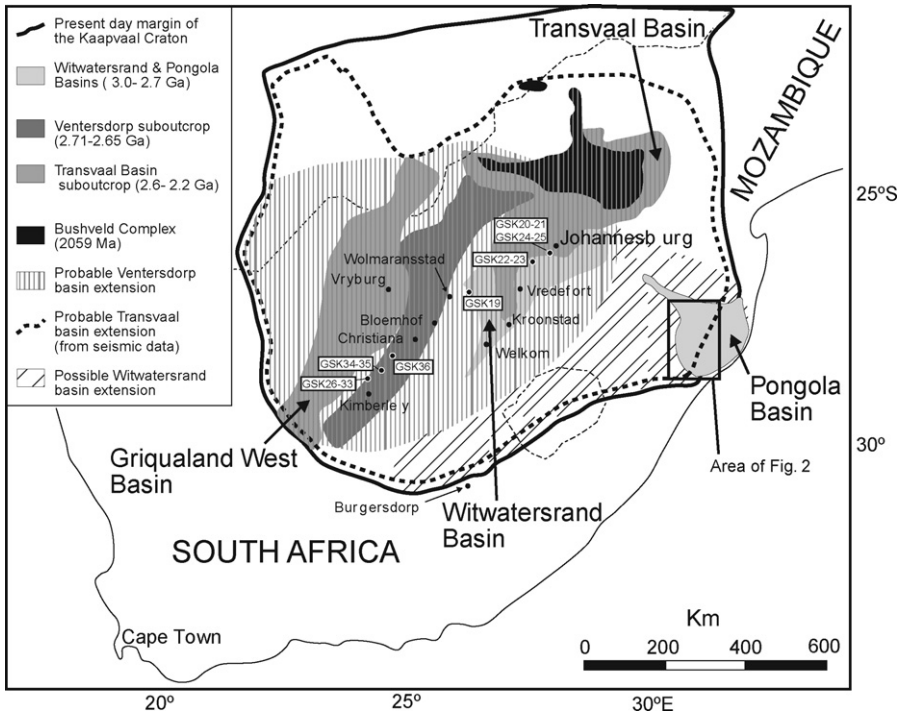


Fig. 1. Simplified outcrop map of the Kaapvaal Craton showing probable basin extensions. Also shown are the Ventersdorp Supergroup sample locations.

2.2. *Ventersdorp Supergroup (ca. 2.71–2.70 Ga)*

The Neoarchaean Ventersdorp Supergroup outcrops over an area of about 3000 km^2 (Winter, 1976) across the central Kaapvaal Craton (Fig. 1). Borehole and seismic data have shown that the extent of the Ventersdorp Supergroup is far greater than what can be deduced from the surface exposure (e.g. Tyler, 1979; Tinker et al., 2002). The Ventersdorp Supergroup overlies the Witwatersrand Supergroup with a (mostly angular) unconformity (e.g. Tankard et al., 1982; Tinker et al., 2002; Winter, 1976). The Ventersdorp Supergroup reaches a total maximum thickness of ca. 8 km, of which 5.1 km exist of subaerial volcanics and ca. 2.9 km of sediments (van der Westhuizen et al., 1991). The Ventersdorp Supergroup has been subdivided in three groups, namely: the subaerial flood basalt dominated Klipriviersberg Group, the lithologically diverse Platberg Group (cf. Table 1), and the siliciclastic and flood basalt dominated Pniel Sequence.

Except for a basal conglomerate, the Klipriviersberg Group contains no sediments, suggesting it formed within a relatively short period, during rapid basalt extrusion (e.g. Eriksson et al., 2002; van der Westhuizen et al., 1991). Formations within the unconfomably overlying volcanic dominanted Platberg Group are all conformable (van der Westhuizen et al., 1991). The Pniel Sequence unconformably overlies the Platberg Group. Rifting most probably continued during the formation of the entire Pniel Sequence (e.g. Eriksson et al., 2002; van der Westhuizen et al., 1991), although Clendenin et al. (1988) suggest that the Allanridge basalts (cf. Table 1) formed during a period of thermal subsidence.

All the rocks of the Ventersdorp Supergroup have been affected by lower greenschist facies metamorphism (e.g. Cornell, 1978; Crow and Condie, 1988); locally, the Klipriviersberg Group has experienced middle to upper greenschist facies at ca. 1950 Ma (Layer et al., 1988b).

3. Methods

Two hundred and fifty-four samples from the Nhlebela Formation (Fm.), the Agatha Fm., the Tobolsk Fm., the Kiphunyawa Fm., Post-Pongola dolerite, the Westonaria Fm. and the Allanridge Fm. were collected (Table 1). Sampling in the field was carried out using a petrol engine drill. Samples were orientated with both a magnetic and sun compass. The sun compass directions were used in the sample analyses, although generally there was no significant difference between

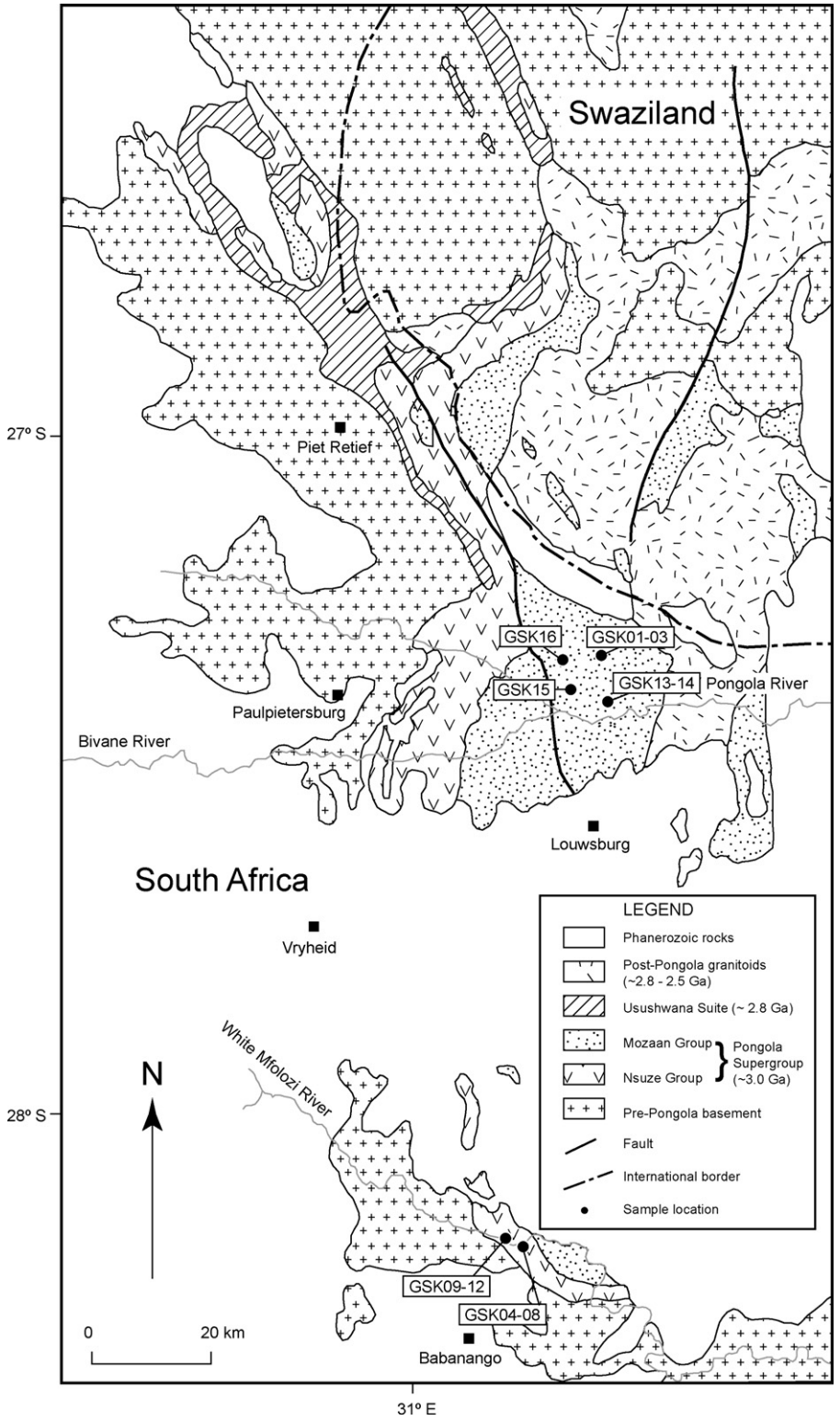


Fig. 2. Simplified geological map of the Pongola Supergroup with sample locations. Modified after Gold and von Veh (1995).

sun and magnetic compass direction. Following earlier studies in the similar-age Pilbara Craton (e.g. Strik et al., 2003), great care has been taken to avoid sites that may have been affected by a lightning induced isothermal remanent magnetisation (IRM). To further test this, 10 oriented hand-specimens were taken from the Westonaria Fm. in the Kloof Gold Mine, Shaft number 4, 41 level, ca. 3160 m below surface. All samples have been carefully checked during demagnetisation and rock magnetic treatment for lightning induced IRM, e.g. by checking for consistency in thermal decay curves and by IRM component analysis. Particular care was taken in determining the paleohorizontal: we measured flow top planes, both by direct readings and by larger scale estimates, i.e. through measurement of lines with different azimuths and dips, from which the bedding plane was reconstructed.

In the laboratory, the hand samples were cored; all cores were cut to standard specimens of 22 mm length. One specimen of each core was thermally demagnetised in a magnetically shielded, laboratory-built furnace. The NRM was measured on a 2 G Enterprises DC-SQUID cryogenic magnetometer. Thermal demagnetisation was done in 16 temperature steps (20, 100, 200, 300, 350, 400, 450, 480, 500, 525, 530, 540, 550, 560, 568, and 575 °C and occasionally 590 °C).

The results of the thermal demagnetisation were plotted on Zijderveld diagrams (Zijderveld, 1967) and principal component analysis (Kirschvink, 1980) was completed to interpret the directions of NRM. Only components with a maximum angular deviation (MAD) less than 10° were accepted for further interpretations.

A modified horizontal translation Curie balance that applies a cycling field (Mullender et al., 1993) and an alternating gradient force magnetometer (AGFM Micro-Mag) were used to measure some magnetic properties and to help identifying the carriers of the NRM of selected samples.

4. Results

The following section describes the results of the thermal demagnetisation of each studied rock unit in sequence from older to younger (Table 1). Apart from an occasional low unblocking temperature present day field component, two main components of magnetisation are present, namely:

- A medium temperature (MT) component that unblocks between 200–350 and 480 °C.
- A high temperature (HT) component that unblocks between 350–500 and 580 °C.

The statistical data of all HT component directions and the data for the MT component directions are given in Tables 2 and 3, respectively.

4.1. Nsuze Group, Nhlebela Fm.

Basalt samples from the Nhlebela Fm. are from fresh exposures in a deeply incised valley where chances of lightning impact were judged to be small. Nevertheless, samples from two out of four sites were found to be severely affected by a lightning induced IRM. The two remaining sites (GSK11–12) have a similar, consistent HT component (Table 2; Figs. 3L–M; 4G). No consistent MT components are present at this location.

4.2. Nsuze Group, Agatha Fm.

Basalts from the Agatha Fm. were sampled in the same valley as those from the Nhlebela Fm., but were not found to be affected by lightning overprints. A consistent HT direction is present in four out of five sites (Table 2; Figs. 3J–K; 4F). A consistent MT component is present in all sites (Table 3; Fig. 4F) that is fully unblocked at ca. 480–500 °C (Fig. 3J–K). A present day field component is sporadically developed (Fig. 4F).

4.3. Mozaan Group, Tobolsk Fm.

Basalts from the Tobolsk Fm. were sampled at two locations and four sites (Fig. 2). A HT component has a consistent direction per site, but site averages are not consistent (Table 2; Fig. 4E; cf. Fig. 3G and H). The HT direction means do not form a confined cluster, either before or after tectonic correction. The MT directions, however, show a consistent direction when no tectonic correction is applied (Table 3; Fig. 4E). A present day field component is not recognised in this formation.

4.4. Mozaan Group, Kiphunyawa Fm.

Diamictite samples from the Kiphunyawa Fm. were taken at two sites, which were sampled to allow for a fold test. GSK13 shows consistent HT, MT and present day field components (Tables 2 and 3; Fig. 4D). Fig. 3F shows a sample in which the MT component is well preserved. Three samples of site GSK14, however, appear to be affected by lightning and the remaining seven have chaotic demagnetisation paths. A fold test could therefore not be carried out.

Table 2
Summary of HT (high temperature) results from the Pongola and Ventersdorp Supergroups

Group	Formation	Site Number	Lithology	Outcrop type	Coordinates (WGS 84)	Strike/dip	N	n	dec°	inc°	k	α95°	R
Karroo age		GSK35	Conglomerate	River incision	S28.32769 E24.71916	subhor.	11	11	131.1	-32.6	1.3	76.2	3.458
Ventersdorp Supergroup													
Pniel	Allanridge	GSK36	Basalt	River incision	S28.09500 E24.86562	subhor.	7	0	n/a	n/a	n/a	n/a	n/a
Pniel	Allanridge	GSK34	Basalt	River incision	S28.32687 E24.71856	subhor.	7	2	195.3	63.2	n/a	n/a	1.998
Pniel	Allanridge	GSK33	Basalt	River incision	S28.46568 E24.37387	163/07 W	7	7	206.1	54.2	28.3	11.5	6.788
Pniel	Allanridge	GSK32	Basalt	River incision	S28.46553 E24.37544	163/07 W	7	4	208.0	54.5	72.5	10.9	3.959
Pniel	Allanridge	GSK31	fissure	River incision	S28.46537 E24.37793	163/07 W	15	0	n/a	n/a	n/a	n/a	n/a
Pniel	Allanridge	GSK30	Basalt	River incision	S28.46524 E24.37793	163/07 W	7	6	195.0	60.2	72,5	7.9	5.931
Pniel	Allanridge	GSK29	Basalt	River incision	S28.50963 E24.39162	subhor.	7	5	173.7	64.6	35.3	13.1	4.887
Pniel	Allanridge	GSK28	Basalt	River incision	S28.51187 E24.39029	subhor.	7	5	180.8	63.7	33.2	13.5	4.879
Pniel	Allanridge	GSK27	Basalt	River incision	S28.51754 E24.38674	subhor.	7	7	164.6	58.7	80.7	6.8	6.926
Pniel	Allanridge	GSK26	Basalt	River incision	S28.51823 E24.38639	subhor.	7	7	178.1	59.6	154.0	4.9	6.961
Pniel	Allanridge	GSK19	Basalt	Road cut	S26.92500 E26.38473	subhor.	7	0	n/a	n/a	n/a	n/a	n/a
Pniel	**Allanridge**						**8**		**188.7**	**60.7**	**82.0**	**6.2**	**7.915**
Klipriviersberg	Westonaria	GSK24	Basalt	Quarry	S26.30111 E27.98459	083/20 S	7	2	124.9	81.4	n/a	n/a	1.984
Klipriviersberg	Westonaria	GSK25	Basalt	Quarry	S26.30055 E27.98294	083/20 S	8	7	78.7	80.9	29.1	11.4	6.794
Klipriviersberg	Westonaria	GSK21	Basalt	Quarry	S26.29844 E27.98459	083/20 S	9	9	40.4	69.8	22.4	11.1	8.643
Klipriviersberg	Westonaria	GSK20	Basalt	Quarry	S26.30021 E27.98314	083/20 S	8	1	62.7	78.1	n/a	n/a	n/a
Klipriviersberg	Westonaria	GSK23	Basalt	Mine	Kloof, shaft #4 41 level	151/30 SW	5	0	n/a	n/a	n/a	n/a	n/a
Klipriviersberg	Westonaria	GSK22	Basalt	Mine	Kloof, shaft #4 41 level	151/30 SW	5	0	n/a	n/a	n/a	n/a	n/a
Klipriviersberg	**Westonaria**						**4**		**66.2**	**79.1**	**84.7**	**10.0**	**3.965**
Pongola Supergroup													
?	Post-Pongola	GSK15	Diorite	River incision	S27.40616 E31.22810	018/45 E	10	8	132.2	32.1	60.7	7.2	7.885
?	**Post-Pongola**						**1**		**132.2**	**32.1**	**60.7**	**7.2**	**7.885**
Mozaan	Kiphunyawa	GSK14	Diamictite	River incision	S27.42398 E31.27691	353/09 E	10	0	n/a	n/a	n/a	n/a	n/a
Mozaan	Kiphunyawa	GSK13	Diamictite	River incision	S27.42163 E31.27467	043/36 SE	10	6	148.1	19.4	101.7	6.7	5.951
Mozaan	**Odwaleni**						**1**		**148.1**	**19.4**	**101.7**	**6.7**	**5.951**
Mozaan	Tobolsk	GSK16	Basalt	Gully	S27.37628 E31.22210	011/21 E	7	5	52.8	-64.8	198.5	5.4	4.980
Mozaan	Tobolsk	GSK03	Basalt	River incision	S27.35403 E31.26595	258/67 N	7	3	119.5	-35.5	642.7	4.9	2.997
Mozaan	Tobolsk	GSK02	Basalt	River incision	S27.35469 E31.26517	279/71 N	7	3	75.2	16.6	67.4	15.1	2.970
Mozaan	Tobolsk	GSK01	Basalt	River incision	S27.35487 E31.26487	251/33 N	7	6	257.4	69.8	71.3	8.0	5.930
Mozaan	**Tobolsk**						**4**		**89.1**	**-9.1**	**1.3**	**n/a**	**1.658**
Nsuze	Agatha	GSK08	Basalt	River incision	S28.23298 E31.18613	318/25 NE	8	8	272.5	51.6	82.1	6.2	7.915
Nsuze	Agatha	GSK07	Basalt	River incision	S28.23304 E31.18602	318/25 NE	7	7	274.0	49.8	180.7	4.5	6.967
Nsuze	Agatha	GSK06	Basalt	River incision	S28.23341 E31.18554	318/25 NE	7	5	273.2	46.8	318.8	4.3	4.987
Nsuze	Agatha	GSK05	Basalt	River incision	S28.23372 E31.18527	318/25 NE	7	1	289.2	57.2	n/a	n/a	n/a
Nsuze	Agatha	GSK04	Basalt	River incision	S28.23267 E31.18661	318/25 NE	7	0	n/a	n/a	n/a	n/a	n/a
Nsuze	**Agatha**						**4**		**276.7**	**51.5**	**157.6**	**7.3**	**3.981**
Nsuze	Nhlebela	GSK09	Basalt	River incision	S28.24441 E31.16114	312/26 NE	8	0	n/a	n/a	n/a	n/a	n/a
Nsuze	Nhlebela	GSK10	Basalt	River incision	S28.24457 E31.16147	312/26 NE	8	0	n/a	n/a	n/a	n/a	n/a
Nsuze	Nhlebela	GSK11	Basalt	River incision	S28.24426 E31.16164	312/26 NE	8	7	317.0	47.1	63.4	7.6	6.905
Nsuze	Nhlebela	GSK12	Basalt	River incision	S28.24385 E31.16154	312/26 NE	8	2	336.4	44.1	n/a	n/a	1.976
Nsuze	**Nhlebela**						**2**		**327.0**	**46.0**	**n/a**	**n/a**	**1.985**

Results are in stratigraphic order. Grey areas show means. Abbreviations: *N* = number of specimens from each site and number of sites contributing to the mean, *n* = number of specimens accepted, dec = declination, inc = inclination, Plat = palaeolatitude, VGP lat (long) = latitude (longitude) of VGP (virtual geomagnetic pole), n/a = not applicable.

Table 3
Summary of MT (medium temperature) results from the Kaapvaal Craton

Group	Formation	Site Number	Lithology	N	n	dec°	inc°	k	α95°	R	Plat°	VGP lat°	VGP long°
Karroo age		GSK35	Conglomerate	11	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Ventersdorp Supergroup													
Pniel	Allanridge	GSK36	Basalt	7	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Pniel	Allanridge	GSK34	Basalt	7	6	339.5	-54.0	71.8	8.0	5.930	34.5	71.5	270.0
Pniel	Allanridge	GSK33	Basalt	7	2	325.0	-64.7	n/a	n/a	1.990	46.6	57.3	251.1
Pniel	Allanridge	GSK32	Basalt	7	7	324.1	-54.0	69.1	7.3	6.913	34.5	59.0	273.9
Pniel	Allanridge	GSK31	fissure	15	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Pniel	Allanridge	GSK30	Basalt	7	4	313.7	-49.3	41.6	14.4	3.928	30.2	49.9	280.3
Pniel	Allanridge	GSK29	Basalt	7	5	345.6	-54.8	47.5	11.2	4.916	35.3	76.0	261.5
Pniel	Allanridge	GSK28	Basalt	7	2	330.2	-51.4	n/a	n/a	1.999	32.1	64.1	279.1
Pniel	Allanridge	GSK27	Basalt	7	3	336.5	-60.3	55.3	16.7	2.964	41.2	67.0	254.6
Pniel	Allanridge	GSK26	Basalt	7	2	342.1	-63.0	n/a	n/a	1.986	44.5	68.6	241.4
Pniel	Allanridge	GSK19	Basalt	7	3	328.0	-38.4	37.2	20.5	2.946	21.6	60.4	299.7
Klipriviersberg	Westonaria	GSK24	Basalt	7	6	339.4	-47.4	60.4	8.7	5.917	28.5	71.6	286.3
Klipriviersberg	Westonaria	GSK25	Basalt	8	6	331.7	-49.8	54.0	9.2	5.907	30.6	64.8	281.5
Klipriviersberg	Westonaria	GSK21	Basalt	9	4	320.8	-51.6	23.2	19.5	3.871	32.2	55.5	278.6
Klipriviersberg	Westonaria	GSK20	Basalt	8	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Klipriviersberg	Westonaria	GSK23	Basalt	5	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Klipriviersberg	Westonaria	GSK22	Basalt	5	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Ventersdorp				127	50	331.0	-53.6	78.7	4.9	11.860	34.1	64.7	272.6
Pongola Supergroup													
?	Post-Pongola	GSK15	Diorite	10	3	333.0	-48.8	227.9	8.2	2.991	29.7	66.2	289.2
Mozaan	Kiphunyawa	GSK14	Diamictite	10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Mozaan	Kiphunyawa	GSK13*	Diamictite	10	7	282.7	-38.9	6.5	25.6	6.080	22.0	20.7	286.5
Mozaan	Tobolsk	GSK16	Basalt	7	6	316.7	-69.5	55.7	9.1	5.910	53.2	49.0	250.0
Mozaan	Tobolsk	GSK03	Basalt	7	7	339.5	-46.0	216.6	4.1	6.972	27.4	71.8	296.5
Mozaan	Tobolsk	GSK02	Basalt	7	8	331.9	-47.7	99.9	5.6	7.930	28.8	65.2	291.3
Mozaan	Tobolsk	GSK01	Basalt	7	4	329.9	-51.6	36.8	15.3	3.919	32.2	63.5	283.3
Nsuze	Agatha	GSK08	Basalt	8	8	334.3	-56.3	155.0	4.5	7.955	36.9	66.8	272.8
Nsuze	Agatha	GSK07	Basalt	7	7	324.3	-55.3	140.4	5.1	6.957	35.8	59.0	277.8
Nsuze	Agatha	GSK06	Basalt	7	7	337.1	-58.1	270.7	3.7	6.978	38.8	68.3	266.2
Nsuze	Agatha	GSK05	Basalt	7	6	328.6	-62.0	507.1	3.0	5.990	43.2	60.7	262.0
Nsuze	Agatha	GSK04	Basalt	7	1	316.8	-37.8	n/a	n/a	1.000	21.2	50.3	301.4
Nsuze	Nhlebela	GSK09	Basalt	8	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Nsuze	Nhlebela	GSK10	Basalt	8	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Nsuze	Nhlebela	GSK11	Basalt	8	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Nsuze	Nhlebela	GSK12	Basalt	8	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Pongola				126	47	328.9	-52.6	57.6	6.4	9.844	33.2	63.1	278.2
All				**34**	**22**	**330.1**	**-53.2**	**70.5**	**3.7**	**21.702**	**33.8**	**64.0**	**275.2**

Consistent MT directions were found only in both the Pongola and the Ventersdorp Supergroups. Results are grouped by formations and supergroups. See Table 2 for additional sample data and explanation of abbreviations. Sites with an asterisk are not included in the mean.

4.5. Post-Pongola dolerite

A fine-grained Post-Pongola dolerite (precise age unknown) was sampled at one site. A consistent HT component is present in 8 out of 10 samples (Table 2, Figs. 3E, 4C). Three samples also show a consistent MT direction (Table 3; Figs. 3E; 4C). No present day field component is recorded.

4.6. Klipriviersberg Group, Westonaria Fm.

Thermal demagnetisation of hand samples from the Westonaria Fm. basalts, taken from the Kloof Gold Mine, resulted in chaotic Zijderveld plots. It is doubtful that the samples have been affected by the elevated temperatures (>50 °C) in the mine, though the shocks from the blasting of the hallways may have affected the NRM. No components from these underground samples could be isolated. Basalt samples taken from a quarry generated better results. Some 60% of these samples have consistent HT directions (Table 2; Fig. 4B) and ca. 40% have consistent MT directions (Table 3; Fig. 4B). The present day field direction is only occasionally present. Zijderveld diagrams show clear progressive demagnetisation paths (Fig. 3C and D).

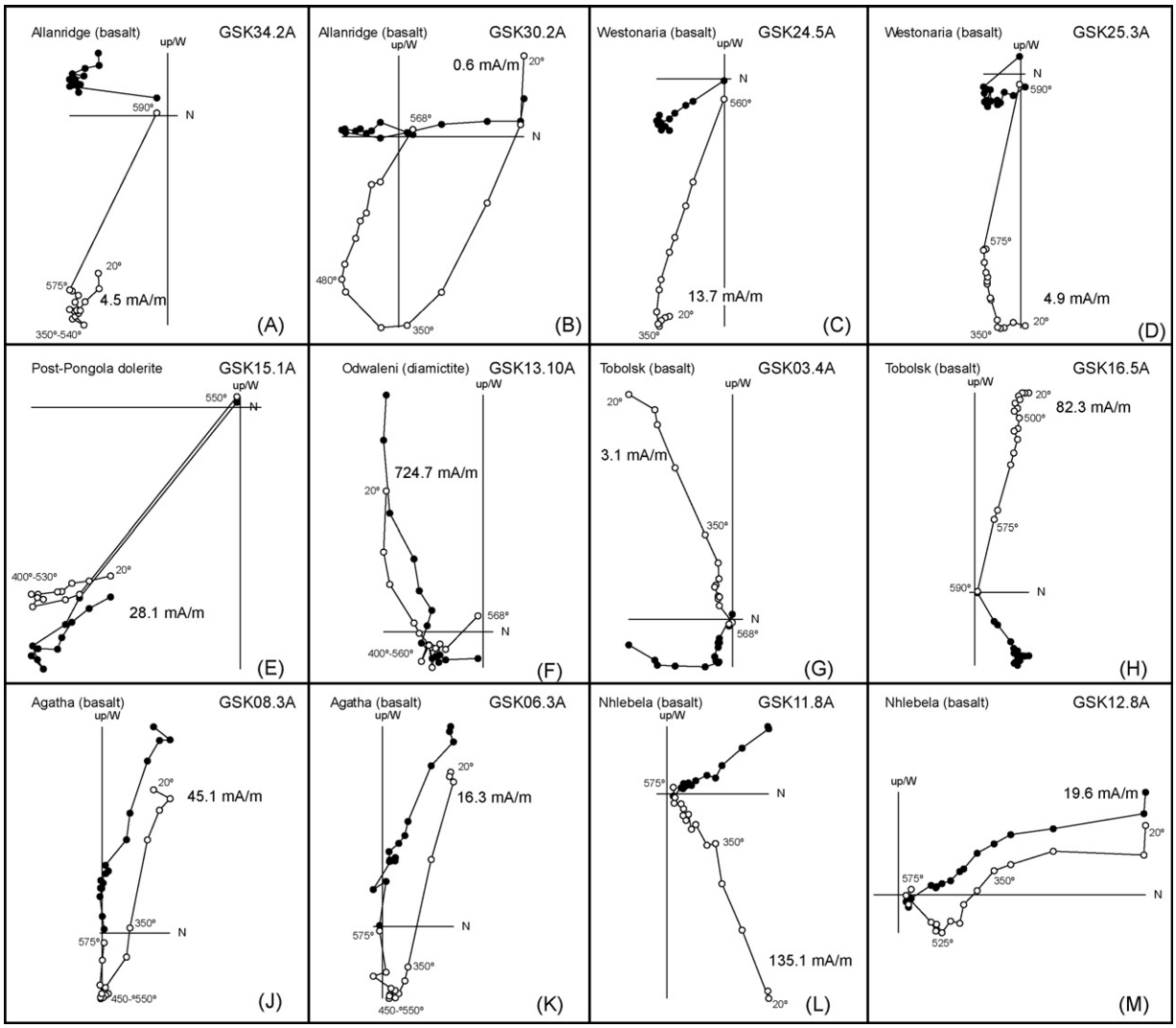


Fig. 3. Selection of characteristic Zijderveld diagrams of samples from all studied formations. Key temperature steps are shown. Specimen numbers indicate location (Figs. 1 and 2, Table 2) followed by sample number.

4.7. Pniel Sequence, Allanridge Fm.

Basalts from the Allanridge Fm. were sampled at four different locations yielding eleven individual sites (cf. Fig. 1; Table 2). At most sites, the quality of results is good (with one exception: site GSK36 was affected by lightning). Site GSK31 was sampled to allow for a "breccia test". Here, a fissure filled with randomly oriented blocks of basalt embedded in a younger basalt matrix allows an experiment similar to a conglomerate test. Random orientations of HT directions of the basalt blocks would indicate that the preserved NRM in sites away from the fissure might have preserved their original NRM. However, Zijderveld diagrams from site GSK31 show a chaotic pattern and no HT components could be isolated, and this "breccia test" is therefore inconclusive. The remaining sites in the Allanridge Fm. show well-developed HT components (Fig. 3A and B) that have a consistent direction (Table 2; Fig. 4A). The MT component is also clearly present (Fig. 3B) and also has a consistent direction (Table 3; Fig. 4A).

4.8. Field tests

Three formations were sampled with the intention to apply a fold test, namely the Tobolsk Fm., the Kiphunyawa Fm., and the Westonaria Fm. Of these, the Tobolsk

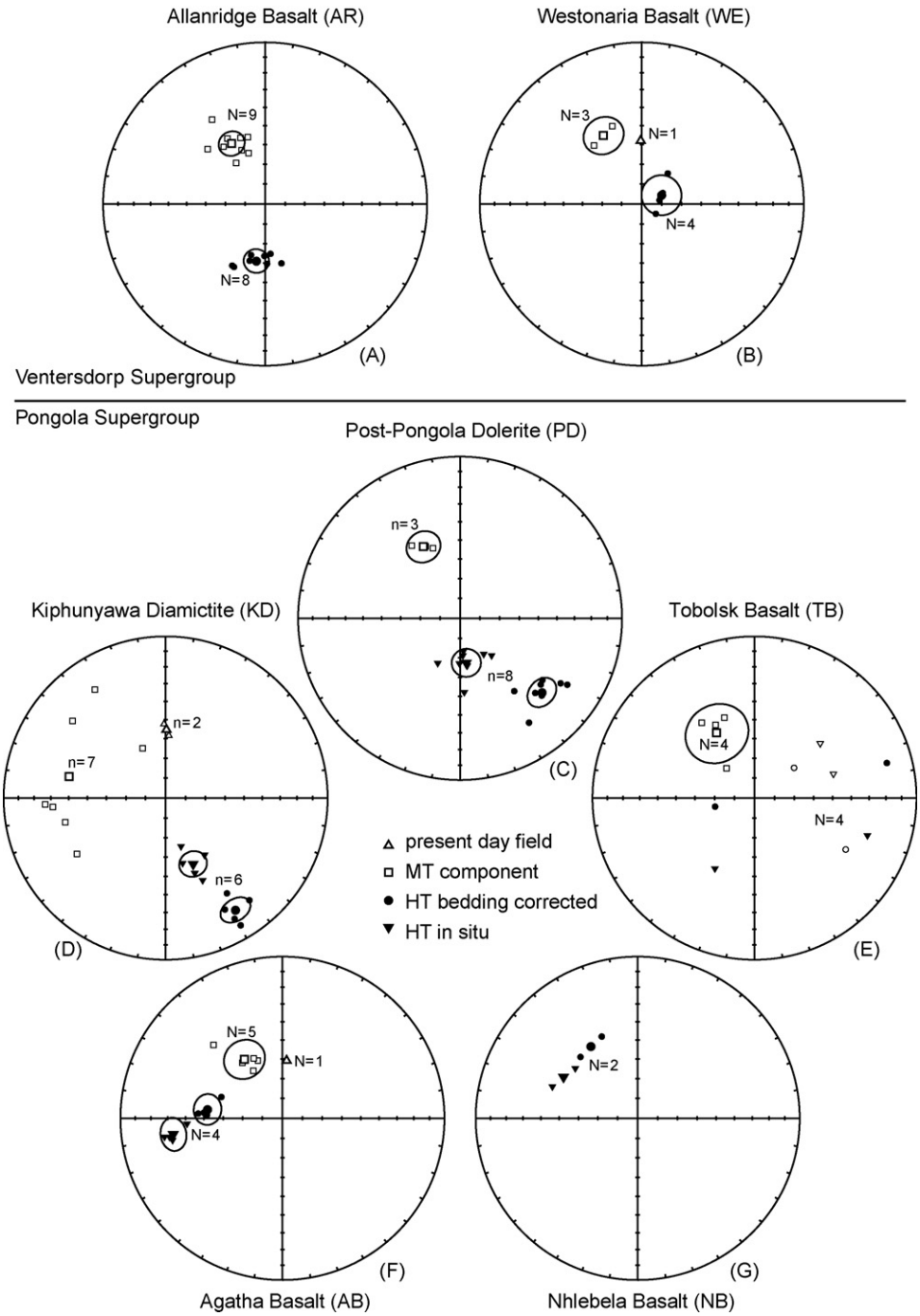


Fig. 4. Equal area plots of results from all studied packages and dykes. (A), (B), (E), (F) and (G) represent equal area plots of multiple sites. N = total number of sites with accepted data, small symbols are site means, larger symbols are the mean of site means with the corresponding α_{95} circles (where applicable). (C) and (D) represent single sites, n = total number of specimens, larger symbols are the site means showing error circles. Open (closed) symbols indicate negative (positive) inclinations, circles are tectonically corrected high temperature (HT) directions, inverted triangles are in situ HT directions, squares represent the secondary (MT) overprint, and triangles are a present day field overprint.

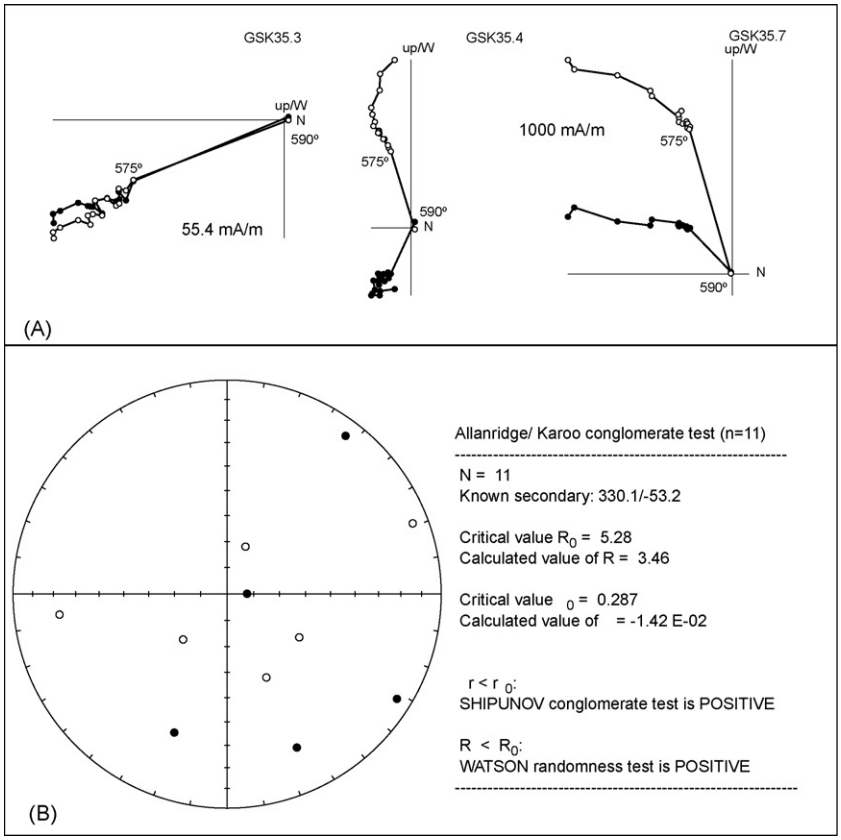


Fig. 5. (A) Zijderveld diagrams of selected conglomerate samples. Unblocking is gradual until 575 °C, followed by rapid unblocking at the next temperature step (590 °C). (B) Equal area plot of HT directions from the conglomerate test, including the statistical data. Both Watson and Shipunov tests are positive. Conventions as in Fig. 4.

Fm. is the only formation that does not have a consistent HT component site mean, either before or after tectonic correction (Fig. 4E). Thus the fold test could not be carried out. Further, one site from the two Kiphunyawa Fm. Sites is affected by lightning induced IRM, which means that here too the fold test could not be completed. Finally, only one of two sample locations in the Westonaria Fm. generated results. Again, this prevented the application of a fold test.

Although the breccia test was inconclusive, lower Permian (295–255 Ma) Karoo Supergroup conglomerates that unconformably overlie the Allanridge Fm. close to the town of Winsorton (Site GSK35, Fig. 1) give excellent results. Eleven boulders of various rock types all have stable demagnetisation paths and unblock at 580 °C (Fig. 5A). The HT directions are plotted in Fig. 5B. The Watson test for uniform randomness (Watson, 1956) gives a positive result when applied to these HT directions. The Shipunov conglomerate test (Shipunov et al., 1998) gives a positive result as well, where the used vector is a known secondary component, namely the mean of all MT directions (cf. Table 3). These results imply that the magnetisation preserved in the Allanridge basalts is at least older than 275 ± 20 Ma. However, the possibility that the preserved NRM is of primary origin cannot be confirmed by field tests at this stage.

4.9. Carriers of natural remanent magnetisation

Progressive thermal demagnetisation (Fig. 3) shows that no remanence remains after heating at 580 °C. Haematite is therefore excluded as carrier of the NRM. Intensity decay curves typically show one or two plateaus (Fig. 6). Where there are two plateaus, the first intensity drop occurs between 200 and 350 °C and the second intensity drop starts at ca. 540 °C, followed by a steep decline to zero at 580 °C. Where there is only one plateau, intensity is stable until ca. 540 °C, after which it steeply declines to zero at 580 °C. The component that

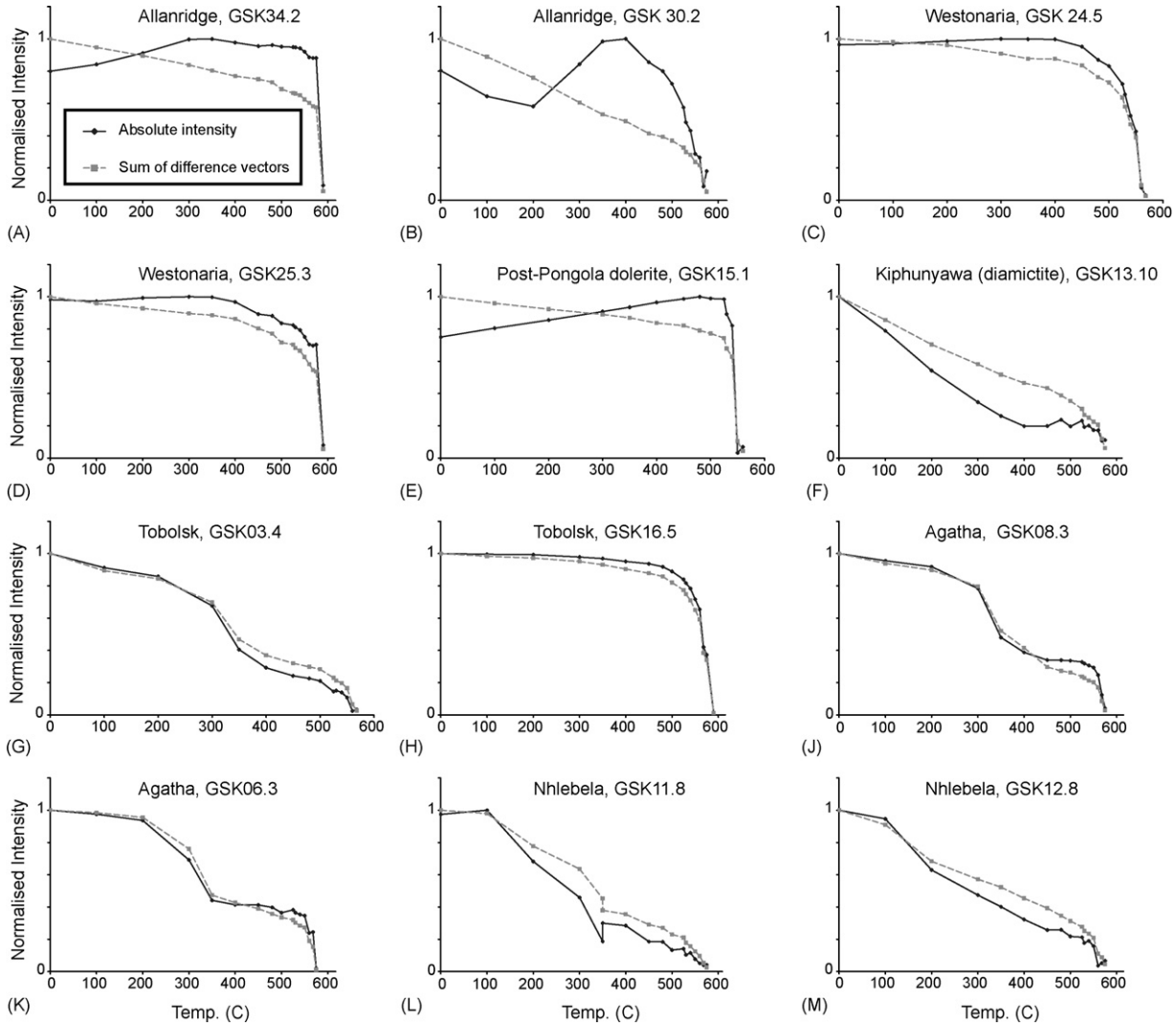


Fig. 6. Normalised magnetic intensity *vs.* temperature. The diagrams correspond to the Zijderveld diagrams of Fig. 3, in the same order. Diagrams show either a fast decrease in intensity, starting at ca. 540 °C (A–F, H) or two decreases, of which one at ca. 300–350 °C and the other at ca. 540 °C (G, J–M). Black graphs represent the absolute intensity decay, grey graphs represent the decay of the sum of difference vectors.

unblocks at 580 °C is single domain (SD) magnetite, and the small unblocking window suggests it is probably very fine-grained. IRM acquisition data (Fig. 7) show a best fit of two components, which have overlapping coercivity spectra. The high coercivities (high Hc) are not indicative of haematite (because samples unblock at 580 °C), but instead suggest slightly oxidised magnetite (van Velzen and Zijderveld, 1995). This could well agree with the decrease in intensity at 350 °C, which may indicate inversion of maghemite (Dankers, 1978).

Experiments with the Curie balance show that the contribution of SD magnetite to the bulk magnetic material of the sample is unusually low, and the samples show mainly paramagnetic behaviour (Fig. 8). The intensity of the NRM in the Kaapvaal specimens is particularly low: on average, the magnetic intensity of the specimens does not exceed 5–10 mA/m, an observation that was also made for Neoarchaean flood basalts of the Pilbara Craton (e.g. Strik et al., 2003). By contrast, in modern basalts such as those from Iceland, magnetic intensities of 10 A/m in standard specimens are common. For the Archaean, samples with high intensities are usually affected by a lightning induced IRM.

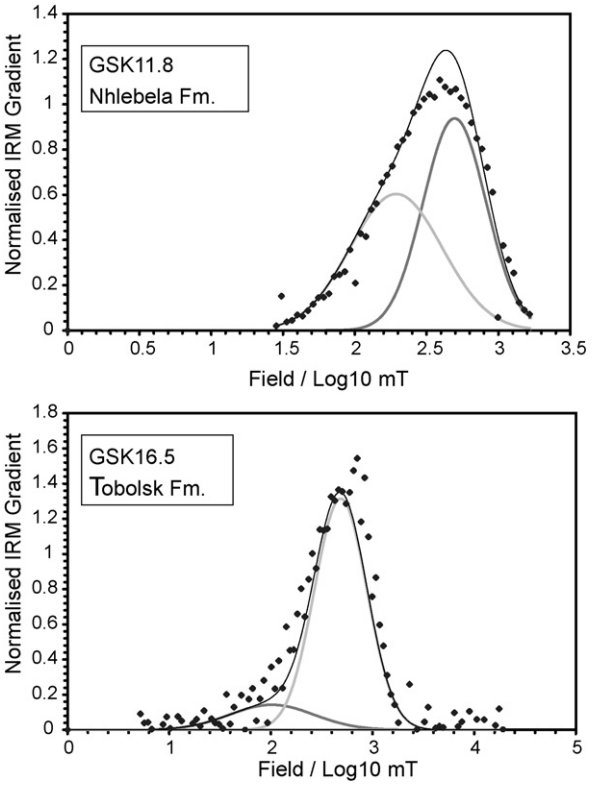


Fig. 7. IRM component analysis as normalised IRM acquisition gradient *vs.* applied field for selected samples. See Fig. 3 for corresponding Zijderveld diagrams of these samples. Most samples are too weakly magnetised to be measured above the noise limits. IRM analysis from the Tobolsk and Nhlebela Formations show a best fit for two overlapping components. Program used was IRMUNMIX V2.2 (Heslop et al., 2002).

5. Discussion and conclusions

5.1. Timing of the MT component magnetisation

From the 34 sites of this study, 23 show internally consistent MT component directions. These 23 MT site mean directions appear to be consistent when no tectonic correction is applied (Table 3). The presence of a ubiquitous common MT component in both the Pongola and Ventersdorp Supergroup samples suggests overprinting by a large-scale and near craton-wide remagnetisation event. This event cannot be related to well-established circum-cratonic orogenic events, since the MT directions are only consistent prior to any tectonic correction. There is significant evidence for near craton-wide thermotectonic events at around 2.0 Ga: the formation of the Bushveld Complex at ca. 2059 ± 1 Ma (U-Pb, titanite, Buick et al., 2001); the 2017 ± 5 Ma Vredefort impact (Reimold et al., 2002); and the 1.87–1.93 Ga basaltic magmatism of the Waterberg-Soutpansberg rifts, part of a greater regional large igneous province of Southern Africa (Hanson et al., 2004). In addition, much of the craton cover may have been effected by ∼1–8–1.9 Ga fluid-induced alteration during the Keiss orogeny along the western margin of the craton (cf. Duana et al., 1988). Further low temperature overprinting event(s) have affected the eastern Kaapvaal Craton as far south as the Barberton Mountain Land, as indicated by $^{40}Ar/^{39}Ar$ biotite dates of ∼2.0 Ga (Layer et al., 1989) and ∼2.0 Ga Rb/Sr dates on carbonates (de Ronde and de Wit, 1994; Weiss and Wasserberg, 1987).

However, the pole position of the MT component of our samples does not coincide with either the palaeopole of the Bushveld Complex (Fig. 9A), or with the less well constrained poles of the Vredefort Dome, the Limpopo direction A, or the in situ Witwatersrand pole (cf. Layer et al., 1988b). This implies that the MT overprint observed in our study has a different origin and is likely of a younger age.

Significant younger candidates for this are the high-grade metamorphic events related to collision of the Natal-Namaqua terrains along the southern and western margins of the Kaapvaal Craton. It is known from thermochronology on lower crustal xenoliths in kimberlites from the Archaean craton, that this orogenesis significantly increased the cratonic geotherm at ∼1.0 Ga across a wide cratonic area. (Schmitz and Bowring, 2004). The palaeopole of Namaqua-Natal (Onstott et al., 1986), however, does not overlap with the MT pole of this study either (Fig. 9A).

A later regional event is the significantly younger emplacement of the Jurassic flood basalts, and massive dolerite dykes and sills of the Karoo Large Igneous Province (LIP) (Erlank, 1984; Hunter and Reid, 1987; Uken and Watkeys, 1997). The bulk of the lavas are preserved in Lesotho, but significant areas in northern South Africa, Zimbabwe, and Namibia were also covered. In addition, the dolerite swarms and thick extensive sills occur throughout Southern Africa (Erlank, 1984). The palaeomagnetic pole of the Karoo basalts (Table 4) and dolerites can be presumed to be primary, based on positive reversal tests (Hargraves and Rehacek, 1997) and is dated at approximately 182 Ma (combined $^{40}Ar/^{39}Ar$, Hargraves and Rehacek, 1997 and U-Pb zircon and baddeleyite, Duncan et al., 1997; Encarnacion et al., 1996). This position overlaps with the pole of the MT component in our Archaean samples (Table 4, Fig 9A). The MT pole is based on 22 sites (Table 3). The near coincidence of the MT and Karoo poles, and the post folding origin of the MT directions, suggests that the Pongola and Ventersdorp Supergroups were partly remagnetised at 180 Ma, during the extensive Karoo igneous event that

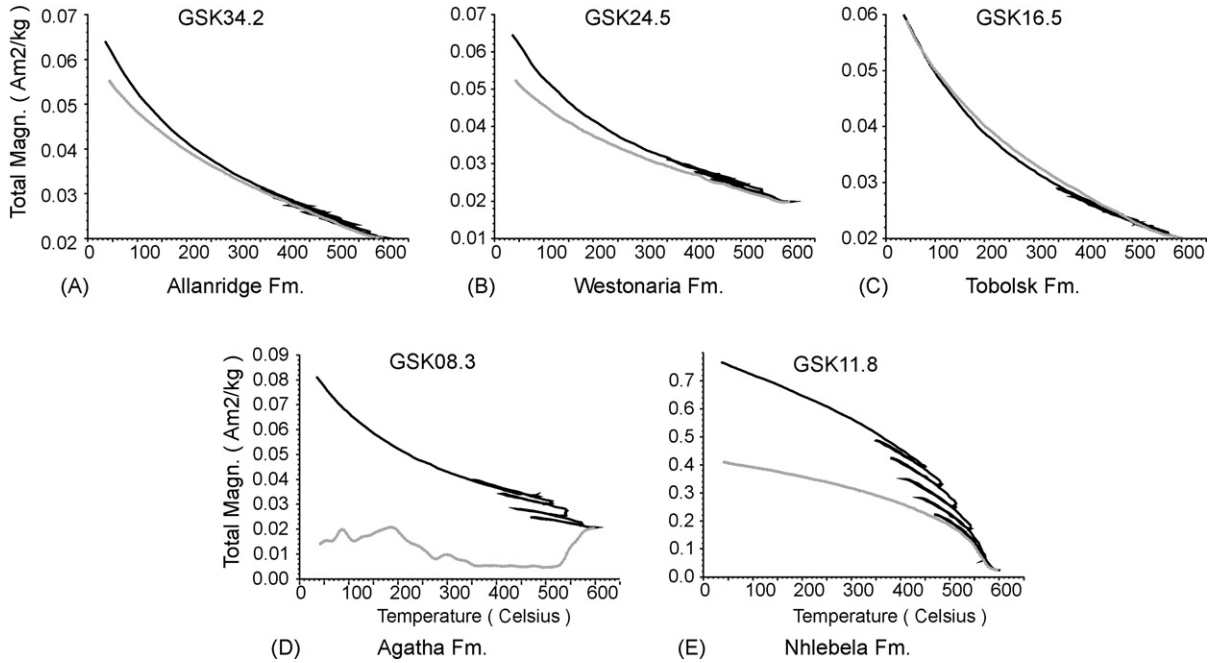


Fig. 8. Total magnetisation *vs.* temperature of selected samples, measured on the Curie balance. Stepwise heating was applied. All steps are in black, except the cooling curve from 600 °C to room temperature, which is drawn in grey. Most samples show pronounced paramagnetic behaviour, except for the Nhlebela Fm. sample. The contribution of (SD) magnetite to the total induced magnetisation appears to be small.

is now generally accepted to be related to the regional mantle hot spot that evolved during the initial break-up of the southern Gondwana supercontinent (e.g. Stern and de Wit, 2004; Storey, 1995; White and McKenzie, 1995). If this interpretation is correct, the thermal pulse related to the Karoo magmatism affected a vast part of the Kaapvaal Craton. Our data indicate that regional remagnetisation during this event has affected at least 240,000 km^2 between Johannesburg, Kimberley and the Pongola Basin. This implies that all palaeomagnetic

Table 4
Summary palaeomagnetic results (this study and published data) for the Kaapvaal Craton between 3.2 and 2.0 Ga

Code	*N*	dec (°)	inc (°)	Plat (°)	latp (°)	longp (°)	α_{95} (°)	Age (Ma)	Ref.
KB	6	156.0	53.0	33.6	69.2	278.3	3.3	183 ± 5	1
MT	20	330.7	−54.0	34.5	64.5	273.6	4.3	183 ± 5	
WD	nya	nya	nya	nya	−15.6	197.1	8.9	1879 − 1872	2
BC	6	195.6	−40.0	22.8	−39.5	227.0	10.9	2050 ± 12	3
OL	20	264.8	−21.1	10.9	0.5	280.7	5.3	2222 ± 13	4
MP	5	65.4	8.9	4.5	19.7	105.7	9.4	2687 ± 6	5, 6
AR	8	188.7	60.7	41.7	−75.0	0.1	8.8	−2708 ± 4	
WE	4	66.2	79.1	68.9	−17.1	47.9	18.8	2714 ± 8	
DB	8	222.7	76.6	64.5	−39.6	4.7	17.5	2782 ± 5	7
MG	10	154.5	85.0	80.1	−32.8	30.8	10.4	>2783	8
UC	18	304.3	52.7	33.3	9.2	347.0	7.6	2875 ± 40	5, 9
PD	1	132.2	32.1	17.4	−45.0	123.6	n/a	Unknown	
KD	1	148.1	19.4	10.0	−55.3	145.2	n/a	Unknown	
AB	4	276.7	51.5	32.2	−9.4	333.0	8.9	2940 ± 22 ?	
NB	2	327.0	46.0	27.4	25.9	358.4	n/a	2984 ± 2.6 ?	
NP	17	293.1	7.0	3.5	17.6	309.8	9.0	3197 ± 18	5, 10

latp = latitude of palaeopole, longp = longitude of palaeopole, see Table 2 for explanation of other abbreviations. Confer Fig. 9 for explanation of codes. Nya = not yet available. Palaeomagnetic references: (1) Hargraves and Rehacek (1997); (2) Hanson et al. (2004); (3) Hattingh (1986); (4) Evans et al. (1997); (5) Layer (1986); (6) Layer et al. (1989); (7) Wingate (1998); (8) Evans and McElhinny (1966); (9) Layer et al. (1988a); (10) Layer et al. (1998).

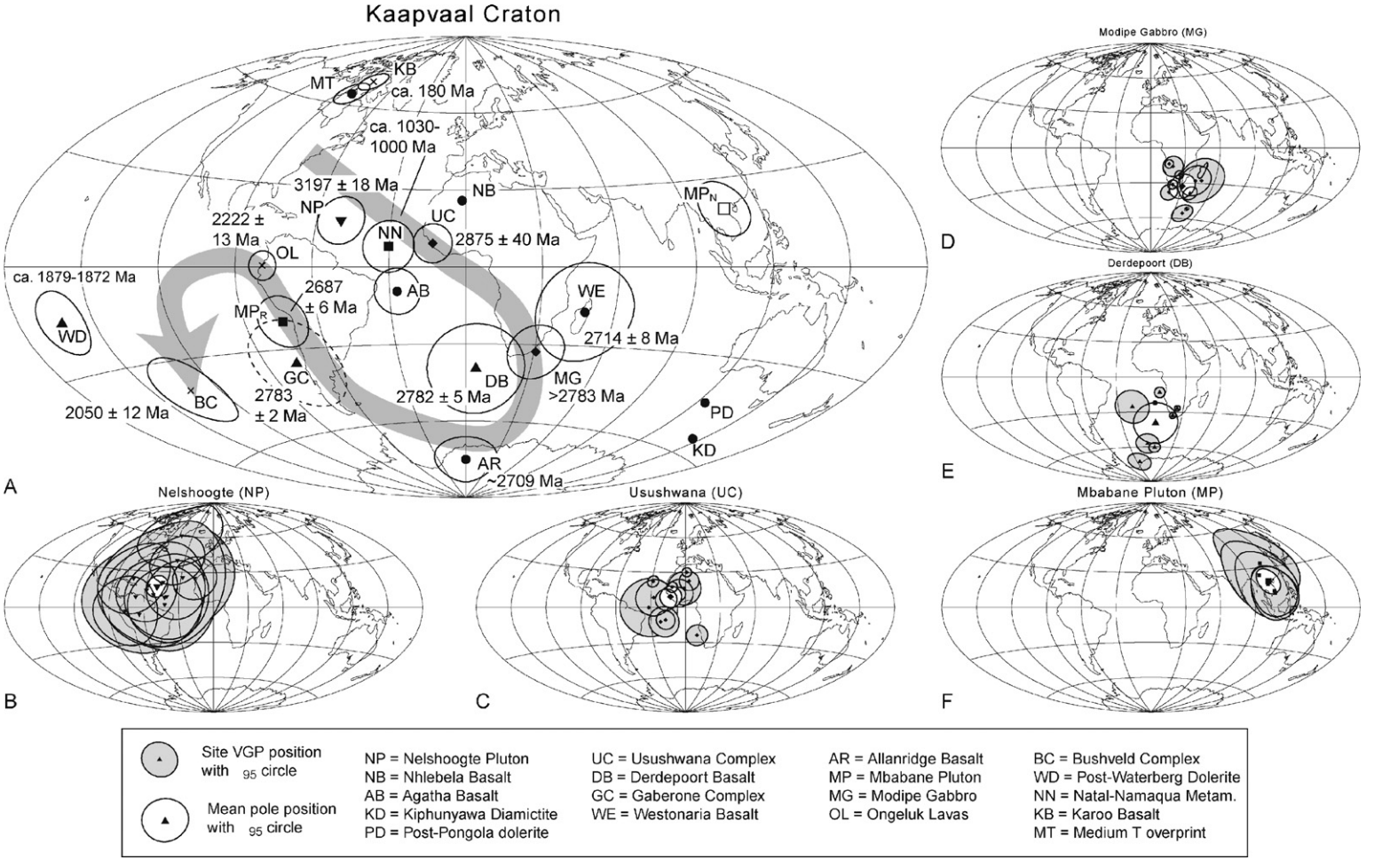


Fig. 9. (A) Aitoff projection of a possible apparent polar wander path (APWP) for the Kaapvaal Craton from ca. 3200 to 1900 Ma (gray arrow) showing approximate pole ages. Bold poles NB, AB, KD, PD, WE, AR and MT from this study. NP (Layer, 1986; Layer et al., 1998); UC (Layer et al., 1988a); GC (Evans, 1966); MG (Evans and McElhinny, 1966); DB (Wingate, 1998); MP (Layer et al., 1989), MP_N is the normal, published direction, MP_R is the reversed direction to allow for a more reasonable APWP; OL (Evans et al., 1997); BC (Hattingh, 1986); NN (Onstott et al., 1986) and KB from (Hargraves and Rehacek, 1997). See legend for code explanation. (B) Nelshoogte Pluton site VGP positions and mean palaeopole (Layer, 1986; Layer et al., 1998). Although the mean pole has a small α_{95}, the VGPs have large errors; therefore, the precision parameter is small ($k = 9$). (C) Modipe Gabbro site VGP position and mean palaeopole (Evans and McElhinny, 1966). The precision parameter is relatively low ($k = 22.4$) and samples have been measured without progressive demagnetisation. (D) Usushwana Complex site VGP position and mean palaeopole (Layer et al., 1988a). The large spreading of the VGP directions result in an inaccurate mean pole determination. The distribution of VGPs is not Fisherian. (E) Derdepoort Basalt site VGP position and mean palaeopole (Wingate, 1998). Site VGPs are often well-determined, but the mean palaeopole underestimates the real distribution. The precision parameter is higher in situ than tilt-corrected. (F) Mbabane Pluton site VGP position and mean palaeopole (Layer et al., 1989), which shows large individual errors.

studies will be hampered by this event, and that possibly some regions or rock types are entirely remagnetised with a Karoo-age overprint.

5.2. Significance of the high temperature (HT) directions

The positive conglomerate test implies that the age of the HT component of the Allanridge Basalt is at least older than ca. 290 Ma. In addition, the existence of a ca. 180 Ma consistent magnetic overprint (MT) suggests that all HT directions from the Pongola and Ventersdorp Supergroups are at least older than 180 Ma. This leaves a gap of ca. 2.5 Gyr during which the HT magnetisations may have been recorded. A case can be made that most HT directions in our samples are pre-folding. The studied sequences do not appear to have been affected by any major tectonic activity after ca. 2000 Ma (cf. for example Eglington and Armstrong, 2004; Layer et al., 1989; Tankard et al., 1982), except for the southern Pongola Basin that was significantly deformed during the Natal-Namaqua orogeny. Indeed, the pole for the Agatha Fm. slightly overlaps with the Natal-Namaqua pole (Fig. 9A) and a possible ca. 1.0 Ga remagnetisation for this formation can therefore not be excluded. The other HT component poles, however, do not coincide with any palaeopoles related to the most obvious regional thermo-tectonic events that could have been the cause of remagnetisation. So, if we assume that the HT component is the result of remagnetisation, it must have occurred prior to ca. 2000 Ma. Furthermore, the remagnetisation event(s) must have been local (as opposed to regional), since all studied formations have different tilt-corrected HT poles.

The palaeopole for the diamictite of the Kiphunyawa Fm. is similar to that of the sampled Post-Pongola dolerite of unknown age. Therefore, it cannot be excluded that the intrusion has remagnetised the diamictite. Further, the Tobolsk lava has consistent HT directions at each site, but the mean of the sites is not consistent, neither pre- nor post-folding. This is indicative of early remagnetisation. The remaining Nhlebela, Westonaria and Allanridge Formations, do not show any direct evidence for remagnetisation and the primary NRM could still preserved in our samples.

5.3. A possible apparent polar wander path (APWP) for the Kaapvaal Craton

Constructing an APWP for the Kaapvaal Craton based on published data for the interval between 3.0 and 1.9 Ga, results in a path defined by eight presumed primary palaeopoles (Fig. 9A). Because of this limited number, a cautious approach must be taken when comparing the four possible primary palaeopoles from our study with this APWP. The palaeopoles from the Nsuze Group, the Nhlebela (NB) and Agatha (AB) Formations plot relatively close to the 2875 ± 40 Ma (Table 1) Usushwana Complex (UC) palaeopole. The palaeopoles of the Westonaria (WE) and Allanridge (AR) Formations plot closest to the 2782 ± 5 Ma Derdepoort Basalt (DB) pole (Table 1; Fig. 9A). If the palaeopoles of the Nhlebela, Agatha, Westonaria and Allanridge Formations are treated as primary, a substantial amount of drift of the Kaapvaal Craton must then have taken place in the Neoarchaean. The palaeolatitude of the Pongola Supergroup averages around 30° latitude, whereas the Ventersdorp palaeolatitude averages around 55°, implying a minimum of 3000 km of displacement in ca. 200 Myr (at an average minimum rate of ca. 1.5 cm yr^{-1}). A similar amount of drift (but at a much greater rate of ca. 100 cm yr^{-1}) has recently been recorded in a palaeomagnetic study of the Neoarchaean of the Pilbara Craton, western Australia. Here, Strik et al. (2003) demonstrated that ca. 3000 km of drift occurred within 3 Myr, between 2721 and 2718 Ma during the formation of the Neoarchaean Nullagine and Mount Jope Supersequences.

When the Post-Pongola dolerite (PD) and the Kiphunyawa Fm. (KD) poles are compared to the above-suggested APWP, it is apparent that they plot far away from the trajectory between the Usushwana Complex (UC) pole and the Derdepoort Basalt (DB) pole (Fig. 9A). If the UC and DB poles are indeed primary, it provides an additional indication that the Kiphunyawa Fm. (KD) is remagnetised by the adjacent Post-Pongola (PD) dolerite. The age of this remagnetisation is not known.

5.4. Assessment of the Kaapvaal Craton APWP

The age of the palaeomagnetic poles – in this study represented by the mean of the virtual geomagnetic poles (VGP) of every individual ChRM direction – are poorly constrained. Other than a positive conglomerate test on a Permian Karoo Supergroup conglomerate, a preserved primary direction cannot be confirmed by robust palaeomagnetic tests. Much of the interpretation of the APWP therefore relies on nine published palaeopoles. Between 3.0 and 1.9 Ga the most important of these poles are: the Usushwana Complex (UC) pole, the Modipe Gabbro (MG) pole, the Derdepoort Basalt (DB) pole and the Mbabane Pluton (MP) pole (Table 4). In addition, the ca. 3.2 Ga Nelshoogte Pluton (NP) pole is listed in the GPMDB, and we tentatively consider it as a starting point

for the APWP of the eastern Kaapvaal Craton. We caution that no solid conclusions can be drawn from the path between 3.2 and 2.9 Ga, since at 3.2 the eastern Kaapvaal Craton had not yet completely amalgamated and stabilised, and stabilisation of the entire craton probably only occurred around 2.9–3.0 Ga (e.g. de Wit et al., 1992; Moser et al., 2001; Schmitz et al., 2004; Schoene and Bowring, 2007). The relevant palaeopoles are discussed below from oldest to youngest.

The palaeopole for the Nelshoogte Pluton is not confirmed by any field tests, and no palaeohorizontal was determined. Although the Nelshoogte data show both normal and reversed directions, the reversal test is indeterminate (Layer, 1986; Layer et al., 1998). With an average α_{95} of 33.7 (average from 14 sites, 3 sites had only one accepted direction), the site VGPs are not well constrained (Fig. 9B). Further, given the wide range of NRM intensities including unusually high intensities (>1.0 A/m), part of the data is likely affected by lightning. The precision parameter for the mean pole position ($k = 9$) questions the statistical relevance of the Nelshoogte Pluton palaeopole, and does not meet the minimum requirements for a reliable palaeomagnetic result (Van der Voo, 1990). The criterion used to accept this palaeopole as a primary position relies only on the fact that the average direction does not coincide with younger palaeopoles. The primary nature of the Nelshoogte Pluton palaeopole can therefore not be verified.

The Usushwana palaeopole cannot be subjected to rigorous palaeomagnetic tests either (Layer et al., 1988a), and no palaeohorizontal can be determined. The Usushwana palaeopole has been inferred as primary because its direction does not coincide with any younger palaeopoles. Although the position is slightly better determined ($k = 22$) than for the Nelshoogte Pluton (Table 4), the spreading of site VGPs is still large (Fig. 9C) and does not conform to a Fisher distribution ($M_e = 1.292$, cf. Fisher, 1953; Tauxe, 1998). This implies that the mean pole has no statistical meaning and can thus not be demonstrated to be primary.

The palaeomagnetic data for the Modipe Gabbro (as well as the adjacent Gaborone Complex) were acquired without progressive demagnetisation, and no palaeohorizontal can be determined. Directions were measured once "stable endpoints" were reached (Evans and McElhinny, 1966) (GC, Fig. 9A, Evans, 1966). The Gaborone Complex pole has a very low precision parameter ($k = 7.04$) that has previously been considered unreliable (Wingate, 1998); and we concur with that. The determination of the Modipe Gabbro pole (Fig. 9D) is slightly better ($k = 22.4$). However, given the used demagnetisation method, the Modipe Gabbro palaeopole cannot be considered much more reliable than the pole for the Gaborone Complex. Also, no field tests can confirm a primary NRM, so this assumption remains untested.

The palaeopole for the Derdepoort Basalt is interpreted as primary based on a positive conglomerate test (Wingate, 1998). However, the site VGPs are dispersed and show a stretched band from the equator to the pole, and there seem to be two distinct groups within the distribution (Fig. 9E). The α_{95} (17.5°) of the mean pole therefore underestimates the real distribution. Further, the precision parameter of the mean pole position decreases from $k = 46$ in situ to $k = 29$ upon tectonic correction. Therefore, the primary nature of the palaeopole for the Derdepoort Basalt cannot be demonstrated convincingly with the available data.

No palaeomagnetic tests are available to test the validity of the primary nature of the Mbabane Pluton pole (Layer et al., 1989) and no palaeohorizontal can be determined. The pole is based on five site VGPs, with an average α_{95} of 23.8° per VGP, which does not result in a well-determined palaeomagnetic pole (Fig. 9F). Individual specimens from the Mbabane Pluton vary in intensity, and many exceed 1.0 A/m (cf. Layer, 1986), which, in our experience, is almost invariably indicative of a lightning induced IRM for these Archaean rocks. Without field tests, the only indication that the Mbabane Pluton pole may be primary is that it does not coincide with younger poles. Hence, there is no solid basis for assuming that this pole is primary.

Palaeoproterozoic poles for the Kaapvaal Craton appear better defined. Those for the Ongeluk lavas (OL), the Bushveld Complex (BC) and the Post-Waterberg dolerites (WD) (Fig. 9A and cf. Table 1) have been successfully subjected to a conglomerate (breccia) test, a fold test, and a reversal test, respectively.

In summary, there appear to be no robust palaeopoles for the Archaean Kaapvaal Craton that can confidently be called primary. The argument that a palaeopole could be primary, because it does not correspond with a younger pole on the APWP, can only rely on a solidly established and detailed APWP. This is not the case yet for the Kaapvaal Craton. Both published data and our new data are of similar variable quality that preclude construction of a reliable Archaean APWP for the Kaapvaal Craton. This implies that estimates of amounts and speed of rifting of the Kaapvaal Craton between 3.0 and 1.9 Ga, as made above and by e.g. Kröner and Layer (1992) and Layer et al. (1989), must be considered with utmost caution.

Also, in comparing the present Archaean APWP of the Kaapvaal Craton to that of other cratons, interpreta-

tions must be considered with great care. Based on the currently available palaeomagnetic data, for example, the Archaean connection between the Kaapvaal and Pilbara Cratons (Strik et al., 2003; Wingate, 1998; Zegers et al., 1998), albeit permissible, has a low level of confidence. Whereas palaeopoles that determine the APWP of the Pilbara Craton have relatively high precision for at least some intervals, this is not yet the case for the Kaapvaal Craton. The large time gaps between palaeopoles of the Kaapvaal Craton therefore still allow for multiple and competing tectonic interpretations.

To conclude, this study has demonstrated the following:

- Five out seven studied formations have provided a mean pole based on a stable HT remanent magnetisation, namely the Nhlebela, Agatha, Westonaria and Allanridge Formations, and a Post-Pongola dolerite. Basalts from the Tobolsk Formation are almost certainly remagnetised, probably early in their tectonic history. The diamictite of the Kiphunyawa Formation was probably remagnetised during the intrusion of a Post-Pongola dolerite of unknown age.
- The age of the characteristic (HT) remanent magnetisation (ChRM) can be constrained to be at least ca. 290 Ma for the Allanridge Fm. based on a positive Karoo Supergroup-aged conglomerate test. For the Nhlebela, Agatha and Westonaria Formations, the ChRM is older than 180 Ma, which is a dated secondary overprint that coincides with the palaeopole of Karoo Supergroup basalts and dolerites and the onset of the break-up of Gondwana.
- There appears to be little effect on the craton of the ca. 1.0 Ga Natal-Namaqua orogeny. The only exception comes from the Agatha Formation that may have been affected by this event. Given that no other major tectonic phase occurred after ca. 2000 Ma and that the palaeopoles of the four formations are all different, the HT magnetisation is likely pre-folding. Because these palaeopoles do not correspond to the poles for the Bushveld Complex or the Vredefort impact, it is possible that the primary NRM is still preserved, although at this time no field tests can confirm this.
- A major part of the Kaapvaal Craton has been affected by a remagnetisation event related to the magmatism of the Karoo LIP. A consistent remagnetisation overprint was found in 20 sites, both in the Pongola and in the Kaapvaal Supergroup. It can be estimated at 180 Ma, the age of the coinciding palaeopole of the Karoo basalts.
- The quality of existing palaeomagnetic data do not allow construction of a reliable Archaean APWP for the Kaapvaal Craton, because no field tests can confirm inferred primary palaeopoles. This implies that available palaeomagnetic data cannot convincingly determine drift rates for the Kaapvaal Craton and cannot be used to test a possible Archaean "*Vaalbara*" connection between the Kaap*vaal* and Pil*bara* Cratons.

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